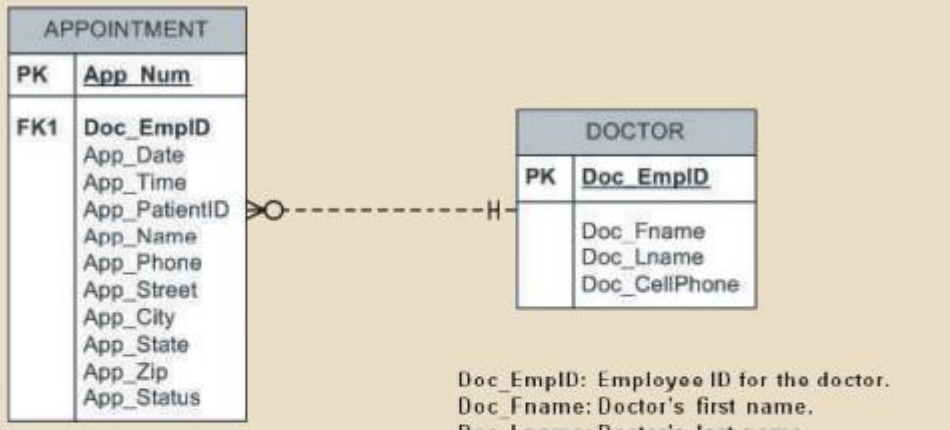
Analyzing Dependency of A Database

Using the descriptions of the attributes given in the figure, convert the ERD shown into a **dependency diagram** that is in at least 3NF.

FIGURE P6.1 APPOINTMENT ERD FOR PROBLEM 1

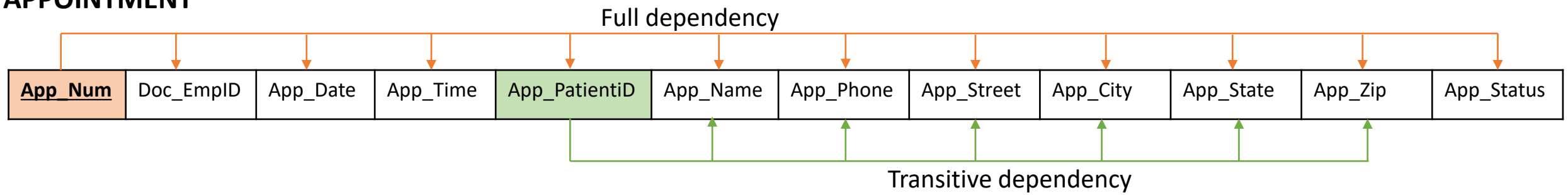


Doc_EmpID: Employee ID for the doctor.
Doc_Fname: Doctor's first name.
Doc_Lname: Doctor's last name.
Doc_CellPhone: Doctor's cell phone number.

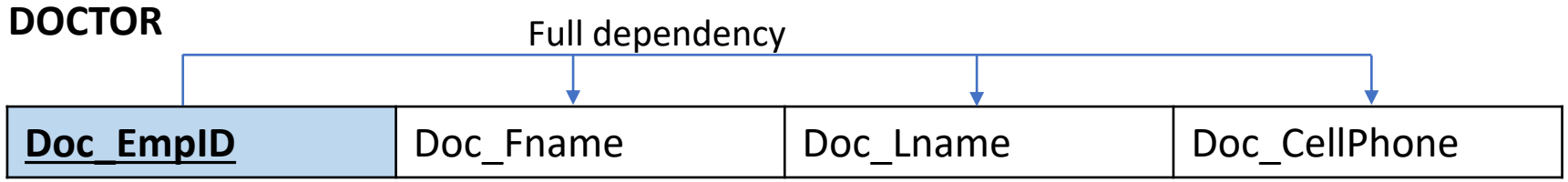
App_Num: System-generated number.
App_Date: The date of the appointment.
App_Time: The time of the appointment.
App_PatientID: The ID number of the patient.
App_Name: The name of the patient.
App_Phone: The contact phone number of the patient.
App_Street: The street address for the patient.
App_City: The city the patient lives in.
App_State: The state the patient lives in.
App_Zip: The zip code for the patient's address.
App_Status: The status of the appointment (pending, closed, cancelled)

Answer:

APPOINTMENT



DOCTOR



ERD Diagram:

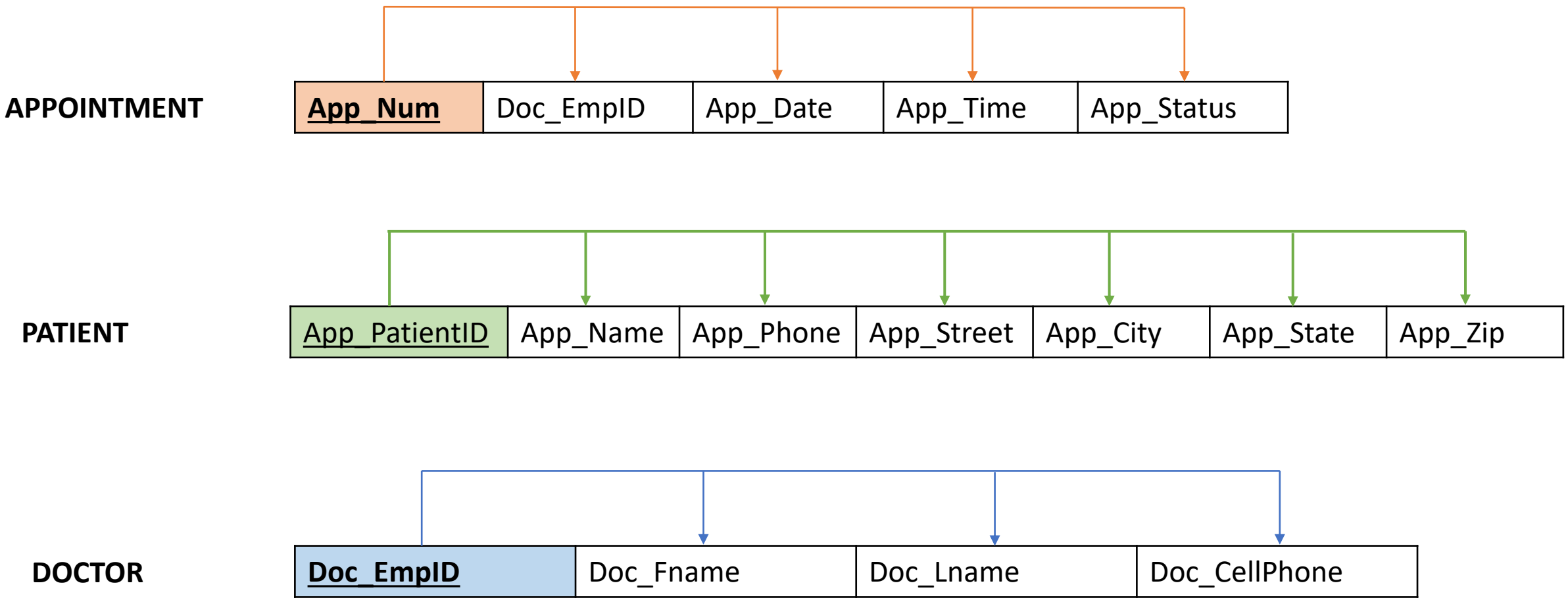
- **APPOINTMENT:** App_Num (PK), Doc_EmpID (FK), App_Date, App_Time, App_PatientID, App_Name, App_Phone, App_Street, App_City, App_State, App_Zip, App_Status
- **DOCTOR:** Doc_EmpID (PK), Doc_Fname, Doc_Lname, Doc_CellPhone

Functional Dependencies (FDs):

- App_Num → Doc_EmpID (FK), App_Date, App_Time, App_PatientID, App_Status
- App_PatientID → App_Name, App_Phone, App_Street, App_City, App_State, App_Zip
- Doc_EmpID → Doc_Fname, Doc_Lname, Doc_CellPhone

Answer & Discussion

Answer: Dependency Diagram (3NF)



Designing A Normalized Database

Using the INVOICE table structure shown below, do the following:

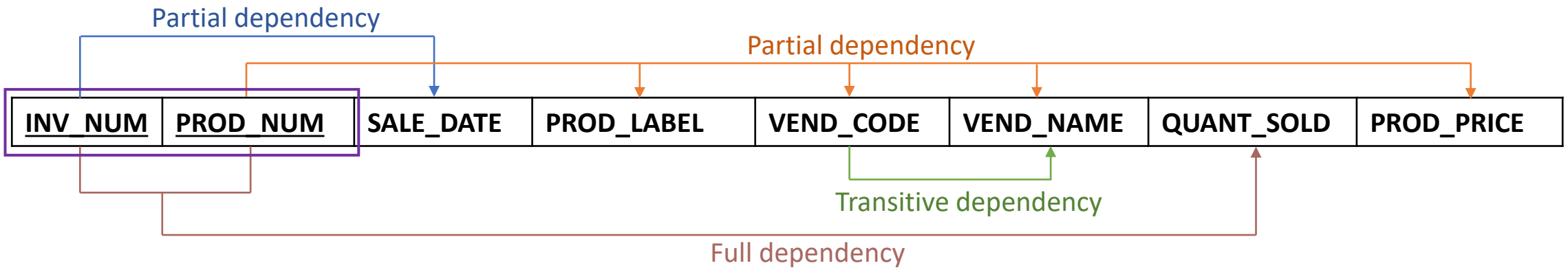
TABLE P6.3

ATTRIBUTE NAME	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE	SAMPLE VALUE
INV_NUM	211347	211347	211347	211348	211349
PROD_NUM	AA-E3422QW	QD-300932X	RU-995748G	AA-E3422QW	GH-778345P
SALE_DATE	15-Jan-2018	15-Jan-2018	15-Jan-2018	15-Jan-2018	16-Jan-2018
PROD_LABEL	Rotary sander	0.25-in. drill bit	Band saw	Rotary sander	Power drill
VEND_CODE	211	211	309	211	157
VEND_NAME	NeverFail, Inc.	NeverFail, Inc.	BeGood, Inc.	NeverFail, Inc.	ToughGo, Inc.
QUANT_SOLD	1	8	1	2	1
PROD_PRICE	$49.95	$3.45	$39.99	$49.95	$87.75

1. Write the **relational schema**, draw its **dependency diagram** and **identify all dependencies**, including all partial and transitive dependencies. You can assume that the table does not contain repeating groups and that any invoice number may reference more than one product. (Hint: This table uses a composite primary key.)

Answer:

Sample	INV_NUM	PROD_NUM	SALE_DATE	PROD_LABEL	VEND_CODE	VEND_NAME	QUANT_SOLD	PROD_PRICE
Sample 1	211347	AA-E3422QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	1	$49.95
Sample 2	211347	QD-300932X	15-Jan-2018	0.25-in. drill bit	211	NeverFail, Inc.	8	$3.45
Sample 3	211347	RU-995748G	15-Jan-2018	Band saw	309	BeGood, Inc.	1	$39.99
Sample 4	211348	AA-E3422QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	2	$49.95
Sample 5	211349	GH-778345P	16-Jan-2018	Power drill	157	ToughGo, Inc.	1	$87.75



Functional Dependencies:

- INV_NUM → SALE_DATE
- PROD_NUM → PROD_LABEL, VEND_CODE, PROD_PRICE
- VEND_CODE → VEND_NAME
- (INV_NUM , PRO_NUM) → QUANT_SOLD

Partial Dependencies:

- SALE_DATE depends only on INV_NUM
- PROD_LABEL, VEND_CODE, PROD_PRICE depends only on PROD_NUM

Full Dependency: QUANT_SOLD depends both on(INV_NUM , PRO_NUM)

Designing A Normalized Database

2. Remove all partial dependencies, write the relational schema, and draw the new dependency diagrams. Identify the normal forms for each table structure you created.

Identify Partial Dependencies

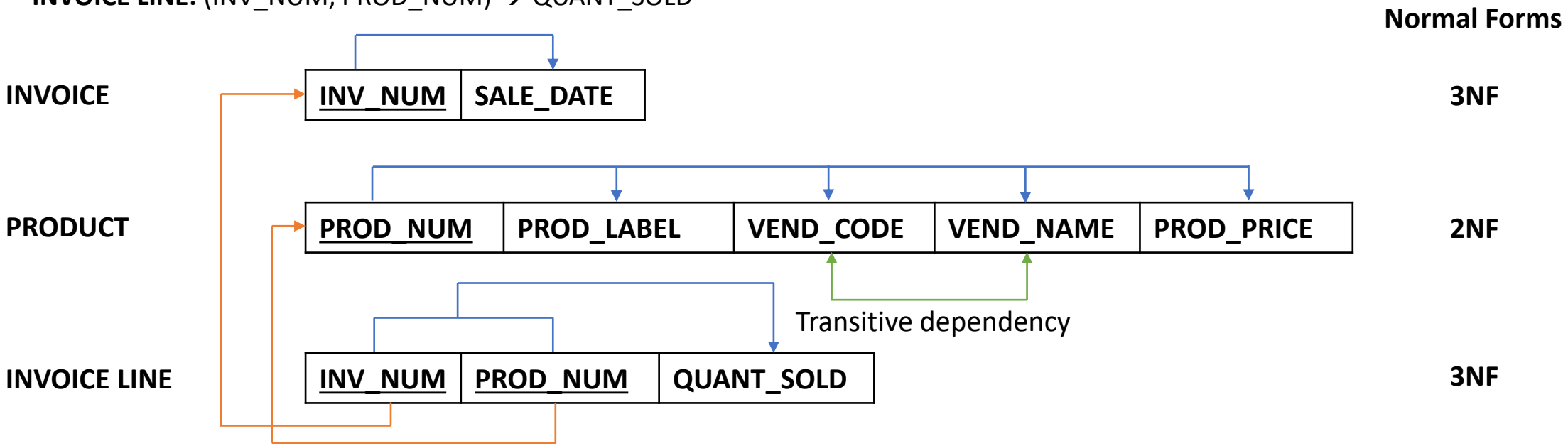
- SALE_DATE depends only on INV_NUM
- PROD_LABEL, VEND_CODE, PROD_PRICE depend only on PROD_NUM

Remove partial dependencies (1NF → 2NF)

- **INVOICE** (INV_NUM, SALE_DATE)
- **PRODUCT** (PROD_NUM, PROD_LABEL, VEND_CODE, VEND_NAME, PROD_PRICE)
- **INVOICE LINE** (INV_NUM, PROD_NUM, QUANT_SOLD, *INV_NUM, PROD_NUM*)

The relational schema:

- **INVOICE:** INV_NUM → SALE_DATE
- **PRODUCT:** PROD_NUM → PROD_LABEL, VEND_CODE, VEND_NAME, PROD_PRICE
- **INVOICE LINE:** (INV_NUM, PROD_NUM) → QUANT_SOLD



Designing A Normalized Database

3. Remove all transitive dependencies, write the relational schema, and draw the new dependency diagrams. Also identify the normal forms for each table structure you created.

Identify Transitive Dependencies

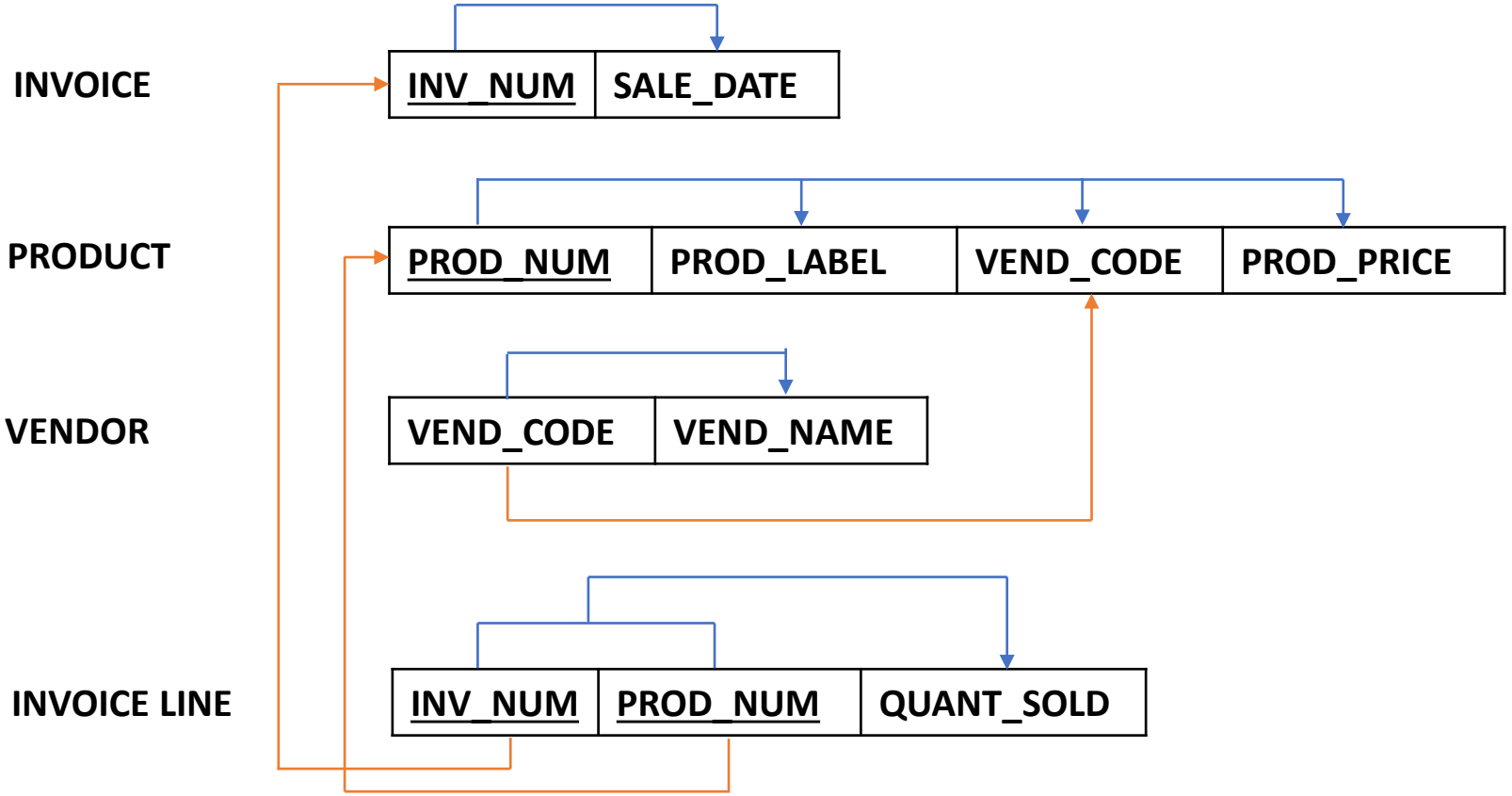
- VEND_CODE → VEND_NAME
- PROD_NUM → VEND_CODE → VEND_NAME

Remove transitive dependencies (2NF → 3NF):

- **PRODUCT** (PROD_NUM, PROD_LABEL, PROD_PRICE, *VEND_CODE*)
- **VENDOR** (VEND_CODE, VEND_NAME)

The relational schema:

- **INVOICE:** INV_NUM → SALE_DATE
- **PRODUCT:** PROD_NUM → PROD_LABEL, PROD_PRICE, *VEND_CODE*
- **VENDOR:** VEND_CODE → VEND_NAME
- **INVOICE LINE:** (INV_NUM, PROD_NUM) → QUANT_SOLD



Normal Forms

3NF

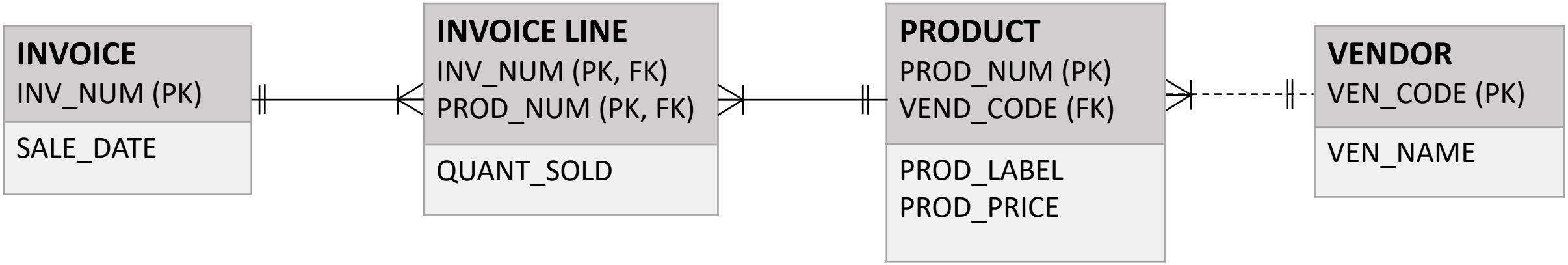
3NF

3NF

3NF

Designing A Normalized Database

4. Draw the Crow's Foot ERD.



Strong entities: INVOICE, PRODUCT, VENDOR
Weak entities: INVOICE LINE

Identifying relationship: INVOICE – INVOICE LINE, PRODUCT – INVOICE LINE
Non-identifying: PRODUCT – VENDOR